

Count Each Person at Their Changing Pace: Multi-Person Tempo-Adaptive Repetition Counting

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Abstract

DRAFT — RESULTS PENDING

*Multi-person repetitive action counting (MRAC) must assign repetitions to the correct person even when several people act asynchronously and each person accelerates, decelerates, pauses, or disappears behind an occluder. The MultiCounter family jointly learns detection, association, and counting from annotated RGB videos, while label-efficient pose counters are primarily developed for a single dominant actor. We study the complementary setting in which a counting model receives identity-indexed pose tracks but uses no person-wise count or period annotation during its intended counting training. Our proposed design shares one pose encoder and a bank of tempo experts across people while keeping temporal state separate for every identity. Within overlapping windows, masked period evidence drives a soft fast-medium-slow router. Expert responses are fused by normalized overlap-add, after which each identity is decoded once, removing the direct arithmetic double-counting path created by summing independent windows. The primary output is a person-wise count vector rather than only a scene total. **Experimental evidence is pending synchronization with the collaborator's frozen implementation and audited MultiRep runs.** The current manuscript is an evidence-gated pre-results specification: exact auxiliary losses, frontend choices, and every empirical statement remain blocked until they are reconciled with frozen code and machine-readable artifacts.*

1. Introduction

Repetitive motion is a useful measurement primitive. Exercise coaching, rehabilitation, industrial monitoring, sports analysis, and human-robot interaction all require more than recognizing that an action repeats: a system must determine how often each participant repeats it. In an uncontrolled scene, however, people need not move together. One person

may slow down while another accelerates, a third may pause, and tracks may contain gaps caused by occlusion. A single response curve for the whole frame can therefore mix identities and incompatible time scales, producing a plausible scene total while assigning the wrong count to each person.

Repetition counting has progressed from online local cycle estimation and motion-frequency analysis [4, 8] to learned temporal self-similarity, density regression, and dynamic-query models [1, 3, 6]. Recent work further addresses exemplar conditioning and multiple action types [9, 15]. These advances largely assume one counted stream. MultiCounter formalizes multi-instance repetitive action counting and jointly predicts people, identities, periodic intervals, and counts [10]; MultiCounter+ strengthens temporal identity consistency and long/short-period awareness [11]. Consequently, multi-person, person-wise, asynchronous, and variable-speed counting are established problems rather than novelty claims of this paper.

The remaining opportunity lies at the intersection of supervision, modality, and temporal granularity. Existing MRAC systems are trained as supervised RGB pipelines, whereas pose-based and label-efficient counting has been developed mostly for a single instance. PoseRAC exploits salient poses in an open-set, zero-shot framework [13]; D²-STX combines RGB and pose streams [12]; and PAMS learns pose representations through period-adaptive multi-scale consistency without repetition annotations [2]. PAMS also evaluates local time warps, but it does not separate simultaneous identities. Conversely, MRAC identity mechanisms do not directly provide an annotation-efficient, pose-driven counter whose tempo assignment can change from window to window along one person's track.

We study that constrained setting. An upstream perception system supplies identity-indexed pose tracks and validity masks. The counting model shares its encoder and tempo experts across people, but maintains independent buffers, routing weights, and output streams for each identity. A masked local period estimate selects a soft mixture of fast, medium, and slow experts within every overlapping window. Instead of counting windows and adding their integer

075 outputs, the model stitches continuous responses with nor-
076 malized overlap-add and performs one final peak extraction
077 per track. This ordering makes the window boundary explicit
078 and removes the direct arithmetic path by which one repeti-
079 tion can be independently counted in two overlapping win-
080 dows; it does not guarantee immunity to phase-misaligned
081 responses.

082 Figure 1 contrasts the target output with a mixed single-
083 stream counter. The formulation does not assume that track-
084 ing errors have been solved: predicted-track experiments
085 must expose their effect, while oracle-track experiments iso-
086 late the counting module. It also does not call the whole
087 system free of all labels. A detector, pose estimator, tracker,
088 or backbone may have external supervision; the narrower
089 intended property is that the counting objective consumes
090 no person-wise repetition count or period labels.

091 The paper makes three falsifiable, currently evidence-
092 gated contributions:

- 093 • We formulate a pose-driven MRAC training setting whose
094 intended counting loss uses no person-wise count or pe-
095 riod annotations, while explicitly accounting for externally
096 supervised perception components. This supervision state-
097 ment must be verified against the frozen training graph
098 before submission.
- 099 • We propose identity-conditioned, window-level tempo
100 routing with shared parameters and per-identity state, fol-
101 lowed by response-domain overlap-add and one track-level
102 decoding pass. Its benefit must be separated from the ef-
103 fects of tracking and the PAMS representation through
104 controlled ablations.
- 105 • We establish an evaluation contract that pairs the official
106 MultiRep counting metrics with oracle/predicted-track pro-
107 tocols, tracking diagnostics, non-stationary-tempo stress
108 tests, and person-count scaling. Every final table cell must
109 be generated from a frozen result manifest.

110 This pre-results draft freezes the scientific story and its
111 evidence requirements; it does not assert that the multi-
112 person modules are already implemented or that the pro-
113 posed method outperforms any baseline.

114 2. Related Work

115 **Single-instance counting and tempo modeling.** Early
116 learned counters estimated a local cycle length in an online
117 window, while motion-centric methods used time-varying
118 flow differentials and wavelets to address non-stationary
119 repetition [4, 8]. RepNet made temporal self-similarity a
120 learned bottleneck and introduced Countix for class-agnostic
121 counting in the wild [1]. TransRAC regresses a density
122 map from multi-scale temporal correlations [3], and De-
123 TRC represents cycles with dynamic action queries under a
124 single-primary-action assumption [6]. ESCounts conditions
125 on exemplars and also learns an exemplar-free representa-

tion [9]; SkimFocusNet combines coarse context with fine-
grained views and studies exemplar-selected multiple action
types [15]. The latter task and its Multi-RepCount bench-
mark differ from MultiRep: they select an action type within
a video rather than assign counts to simultaneous people.

Pose-based and label-efficient counting. Pose represen-
tations reduce appearance dependence and expose articu-
lated motion directly. PoseRAC models actions through
salient poses, uses foundation generative models to synthe-
size pose-conditioned training data, and incorporates CLIP
text embeddings for open-set counting, enabling zero-shot
prediction without real benchmark training samples [13];
zero-shot prediction should not be conflated with a self-
supervised training objective. D²-STX instead fuses RGB
and pose through decoupled spatial and temporal cross-
attention [12]. PAMS learns from unlabeled skeleton se-
quences using period-adaptive multi-scale temporal consis-
tency and combines inference-time counting experts [2]. Its
expert consensus is not a learned fast-medium-slow mixture,
and its single-stream formulation does not preserve multi-
ple identities. Our intended use of pose and PAMS-TCC is
therefore a foundation, not a novelty claim; the proposed
distinction is identity-conditioned local routing in MRAC
under a no-count-label objective.

Multi-instance repetition counting. MultiCounter de-
fines MRAC and introduces MultiRep, learning detection,
association, period localization, and person-wise counting
jointly [10]. MultiCounter+ improves robustness and ef-
ficiency with a dynamically updated Siamese query and
anchor-guided similarity reconstruction that represents long
and short periods [11]. These systems already address asyn-
chronous people, variable speeds, pauses, and identity conti-
nuity. Our proposed setting differs along a narrower axis: an
upstream tracker supplies masked pose tracks, the counting
objective is intended to avoid person-wise count and period
labels, and tempo specialization varies across local windows
of each identity. This modular design is not described as
end-to-end, and any comparison to native MRAC systems
must disclose the different input and supervision protocols.

165 3. Method

Status boundary. This section is a provisional method
contract for synchronizing the paper with the collaborator’s
frozen implementation. It defines the intended data flow and
the boundary between upstream perception and counting.
Components marked **SYNC-REQUIRED** are not asserted
as implemented or validated. If the frozen code differs from
this contract, the equations and prose must change to match
the code.

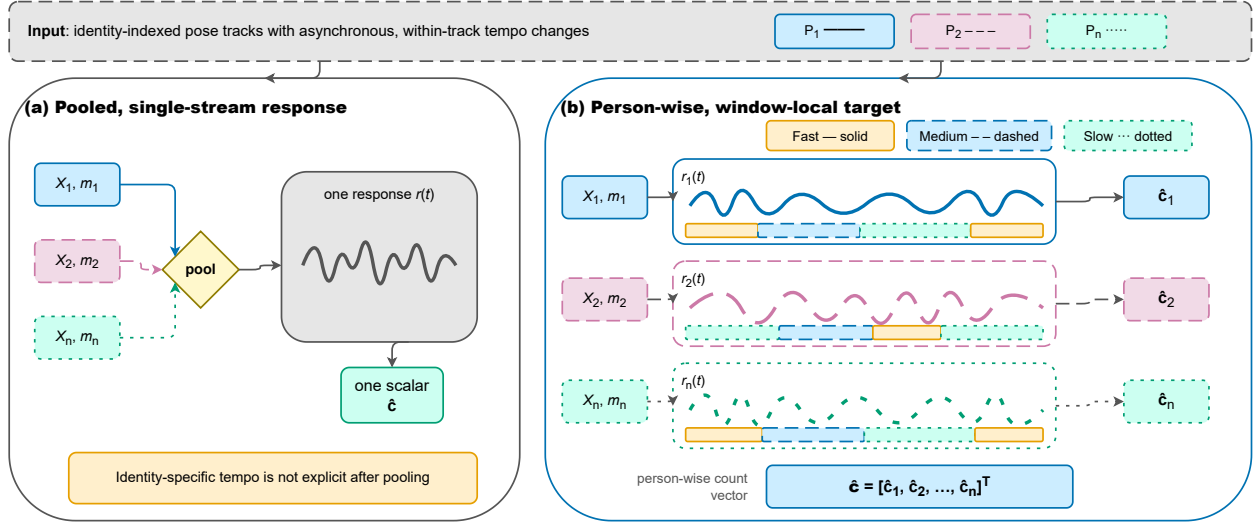


Figure 1. **Target task and output.** A frame-level response can mix people who move at different and changing rates (left). We instead retain identity-indexed masked pose tracks, estimate tempo locally within each track, and return a person-wise count vector (right). The scene total is only a derived diagnostic. This is a problem illustration, not result evidence.

3.1. Problem Formulation and Overview

Given a video, an upstream detector, pose estimator, and multi-object tracker produce N identity-indexed pose tracks. Track i consists of joint coordinates and a validity mask

$$\begin{aligned} \mathbf{X}_i &= [\mathbf{x}_{i1}, \dots, \mathbf{x}_{iT_i}] \in \mathbb{R}^{T_i \times J \times C} \\ \mathbf{m}_i &= [m_{i1}, \dots, m_{iT_i}] \in \{0, 1\}^{T_i} \end{aligned} \quad (1)$$

Here T_i is the temporal extent, J the joint count, C the coordinate dimension, and $m_{it} = 1$ denotes a usable pose at time t . Missing observations keep their original time indices and are masked rather than deleted. An upstream frontend may additionally expose joint-level visibility or confidence, but its deterministic reduction to m_{it} must be recorded in the synchronized preprocessing contract. The primary prediction is

$$\hat{\mathbf{c}} = [\hat{c}_1, \dots, \hat{c}_N]^T \in \mathbb{N}_0^N \quad (2)$$

The optional total $\sum_i \hat{c}_i$ is a diagnostic rather than the task definition. The counter cannot undo an identity switch already embedded in its input; Section 4 therefore separates oracle and predicted tracks.

The design in Figure 2 uses a shared pose encoder and a shared bank of fast, medium, and slow experts. “Person-specific” means that observations, masks, temporal buffers, routing variables, and outputs are indexed by identity; it does not mean that a separate parameter set is instantiated for every person. Each track is processed in overlapping windows, whose continuous responses are fused before any discrete counting decision.

3.2. Identity-Conditioned Local Encoding

A single track can accelerate, decelerate, pause, or contain observation gaps, so we do not assign one speed to the whole person. Let L and S denote window length and stride, and let $s_{i\ell}$ be the start of window ℓ . Its local input at $u \in \{1, \dots, L\}$ is

$$\mathbf{x}_{i\ell u} = \mathbf{x}_{i, s_{i\ell} + u - 1}, \quad m_{i\ell u} = m_{i, s_{i\ell} + u - 1} \quad (3)$$

Out-of-range positions are padded with zero mask values. A shared encoder E_θ maps each masked window to embeddings

$$\mathbf{Z}_{i\ell} = E_\theta(\mathbf{X}_{i\ell}, \mathbf{m}_{i\ell}) = [\mathbf{z}_{i\ell 1}, \dots, \mathbf{z}_{i\ell L}] \in \mathbb{R}^{L \times d} \quad (4)$$

Invalid positions are excluded from aggregation, period evidence, and counting responses. Independent application of the same encoder places people in one representation space without pooling their temporal state; d denotes the embedding dimension.

3.3. Local Period Evidence and Soft Routing

We derive routing evidence within each window. With a fixed numerical stabilizer $\epsilon > 0$, the valid-frame mean is

$$\bar{\mathbf{z}}_{i\ell} = \frac{\sum_{u=1}^L m_{i\ell u} \mathbf{z}_{i\ell u}}{\sum_{u=1}^L m_{i\ell u} + \epsilon} \quad (5)$$

We then define a mask-normalized, non-circular autocorrelation at lag δ

$$A_{i\ell}(\delta) = \frac{\sum_{u=1}^{L-\delta} m_{i\ell u} m_{i\ell, u+\delta} \langle \mathbf{z}_{i\ell u} - \bar{\mathbf{z}}_{i\ell}, \mathbf{z}_{i\ell, u+\delta} - \bar{\mathbf{z}}_{i\ell} \rangle}{\sum_{u=1}^{L-\delta} m_{i\ell u} m_{i\ell, u+\delta} + \epsilon} \quad (6)$$

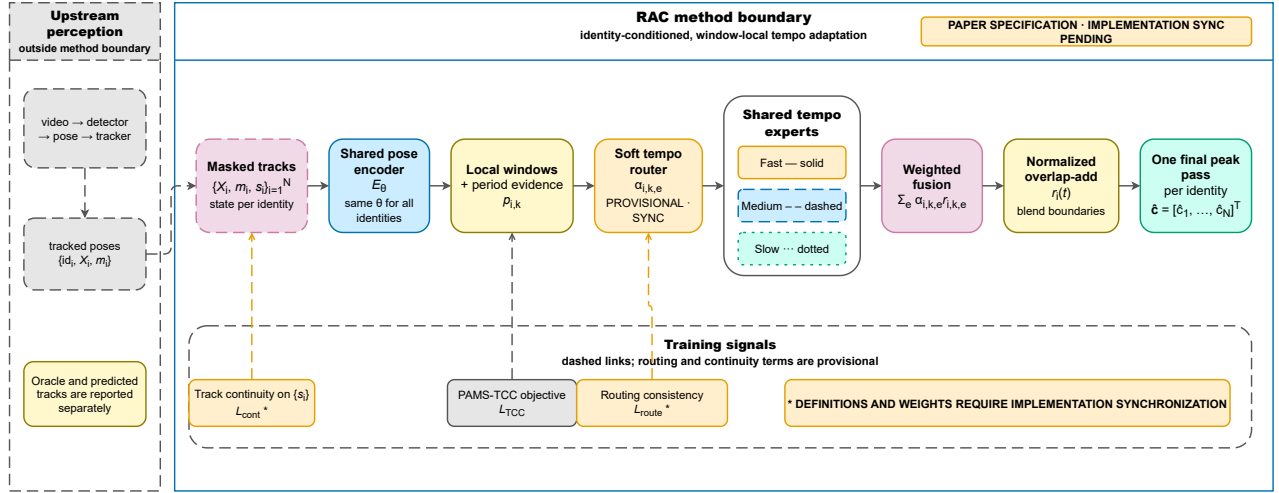


Figure 2. **Provisional method contract.** The upstream detector, pose estimator, and tracker are outside the counting boundary. Within it, parameters are shared across identities while masks, state, local tempo, and output responses remain identity-indexed. Window responses are fused by normalized overlap-add before one final decoding pass per track. Dashed elements require synchronization with frozen training code.

The lag set and minimum number of valid frame pairs are fixed by the protocol; unsupported lags are excluded rather than filled with zeros. For a deterministic taper h , let $\mathcal{F}$ be the nonzero frequency bins in cycles per frame whose reciprocal periods lie inside the declared range. We use the magnitude of the tapered autocorrelation transform as a routing score, not as a calibrated physical power spectral density

$$P_{i\ell}(f) = |\text{FFT}(h \odot A_{i\ell})_f|$$

$$f_{i\ell}^* = \arg \max_{f \in \mathcal{F}} P_{i\ell}(f), \quad \hat{p}_{i\ell} = 1/f_{i\ell}^* \quad (7)$$

$$\gamma_{i\ell} = \frac{P_{i\ell}(f_{i\ell}^*)}{\sum_{f \in \mathcal{F}} P_{i\ell}(f) + \epsilon} \quad (8)$$

Here $\hat{p}_{i\ell}$ is measured in frames and is converted to seconds only with the manifest-bound frame rate. The contract stops the direct gradient from this evidence branch into the encoder; encoder updates can still change routing evidence on later optimization steps. A router maps the descriptor $\mathbf{q}_{i\ell} = [\log \hat{p}_{i\ell}, \gamma_{i\ell}]^\top$ to a probability simplex over shared experts $\mathcal{K} = \{f, m, s\}$. Let π_0 be a fixed fallback distribution used when valid support or spectral confidence is insufficient

$$\tilde{\mathbf{g}}_{i\ell} = \text{softmax}(R_\psi(\mathbf{q}_{i\ell}))$$

$$\mathbf{g}_{i\ell} = \gamma_{i\ell} \tilde{\mathbf{g}}_{i\ell} + (1 - \gamma_{i\ell}) \pi_0 \quad (9)$$

$$\sum_{k \in \mathcal{K}} g_{i\ell k} = 1$$

Because $\mathbf{g}_{i\ell}$ is local, it may change between adjacent windows of one identity. Soft assignment also lets an ambiguous

transition combine neighboring experts rather than commit to a hard tempo bin. The minimum support, confidence calibration, and whether π_0 is uniform or period-conditioned are synchronization-required. Expert-use histograms, entropy, and specialization curves must diagnose routing collapse.

3.4. Expert Fusion and Boundary Handling

Each expert F_{ϕ_k} predicts a nonnegative response over one masked window. Parameters ϕ_k are shared across identities, while each response remains identity-indexed

$$\mathbf{r}_{i\ell}^{(k)} = F_{\phi_k}(\mathbf{Z}_{i\ell}, \mathbf{m}_{i\ell}, \hat{p}_{i\ell}) \in \mathbb{R}_{\geq 0}^L, \quad r_{i\ell u} = m_{i\ell u} \sum_{k \in \mathcal{K}} g_{i\ell k} r_{i\ell u}^{(k)} \quad (10)$$

Summing integer window counts creates a direct arithmetic double-counting path for cycles that straddle an overlap. We instead reconstruct a single response for each identity. With a taper $a_u \geq a_{\min} > 0$ on every covered frame and $\Lambda_i(t) = \{(\ell, u) : t = s_{i\ell} + u - 1\}$, normalized overlap-add gives

$$R_{it} = m_{it} \frac{\sum_{(\ell, u) \in \Lambda_i(t)} a_u r_{i\ell u}}{\sum_{(\ell, u) \in \Lambda_i(t)} a_u m_{i\ell u} + \epsilon} \quad (11)$$

The normalization removes dependence on the number of windows covering an interior frame and excludes padding from the denominator. Only then does a track-level decoder $\mathcal{D}$ extract repetition events

$$\hat{\mathcal{P}}_i = \mathcal{D}(\mathbf{R}_i, \mathbf{m}_i, \{\hat{p}_{i\ell}, \gamma_{i\ell}\}_\ell), \quad \hat{c}_i = |\hat{\mathcal{P}}_i| \quad (12)$$

Thus overlapping windows never contribute separately rounded counts. Fusion does not by itself guarantee a single peak when window responses are phase-misaligned;

boundary-straddling tests must measure whether the fused decoder reduces such errors. The frozen implementation must determine the taper, coverage rule, peak threshold, period aggregation, and minimum-distance rule.

For Period-mAP evaluation, the event set must expose more than its cardinality. The synchronized adapter must export each proposal as

$$\hat{e}_{ij} = (i, \hat{\tau}_{ij}^s, \hat{\tau}_{ij}^e, \hat{s}_{ij}, \hat{B}_{ij}) \quad (13)$$

where the tuple contains its track identity, temporal bounds, confidence, and the spatial/track support required by the frozen official evaluator. The count remains $\hat{c}_i = |\hat{\mathcal{P}}_i|$. Exact interval construction, score calibration, and evaluator matching remain blocked until that adapter is bound to code and tests.

3.5. Training and Synchronization Boundary

The intended representation objective builds on PAMS-TCC [2], which uses temporally adjacent and cyclically corresponding positives across temporal scales, together with negatives from other temporal locations, other videos, and different clusters. At the method-contract level, training is represented by

$$\mathcal{L} = \mathcal{L}_{\text{PAMS-TCC}} + \lambda_{\text{route}} \mathcal{L}_{\text{route}} + \lambda_{\text{track}} \mathcal{L}_{\text{track}} \quad (14)$$

The reserved routing term should transform assignments equivariantly under declared tempo changes rather than force invariance to them. By default, the reserved track term may connect observations only across gaps that retain one supplied track identity. It does not join fragments carrying different tracker IDs; any such re-association would be a separate module with its own signal, supervision disclosure, and evaluation. These are responsibilities, not recovered loss definitions.

SYNC-REQUIRED. Equation 14 is not yet a complete trainable algorithm: it does not specify what trains expert responses or the decoder, how experts specialize, how router collapse is prevented, or how scores are calibrated. Synchronization must recover the exact architectures, response targets or self-supervised signals, pseudo-target provenance, formulas, sample construction, weights, gradient paths, frontend, track normalization, period range, window schedule, fallback route, and decoding settings. Every use of count, period, or identity annotations in training, tuning, calibration, and checkpoint selection must be disclosed. No contribution or result claim depending on these items is admissible before source anchors, tests, and ablation evidence exist.

4. Experiments

Pre-results status. This section fixes the evaluation questions, protocols, and table schemas; it contains no new-method measurement or empirical conclusion. Red entries

are controlled evidence fields. Historical UCFRep dev84 diagnostics, proxy readouts, and runs involving an untrained random Period Head are ineligible for every table below.

4.1. Evaluation Contract

Data and integrity. The primary benchmark will be the exact MultiRep release **PENDING** under its official **PENDING** protocol. The original MultiCounter paper specifies 1,157 synthetic videos, 52,590 periodic events, and a 7:2:1 train/validation/test split, while the current MultiCounter+ repository reports the same size and provides a download without a versioned release identifier or checksums [10, 11]. The final manifest must therefore bind the release identifier, download source and time, video/annotation/split SHA-256 digests, evaluator revision, and preprocessing fingerprint. A run is table-ineligible if any binding is missing or if incompatible artifacts are mixed.

Oracle and predicted tracks. The oracle-track protocol supplies official person identities and spatial support, isolating repetition counting from detection and association. The predicted-track protocol uses one frozen detector-pose-tracker frontend for all compatible track-conditioned counters. Native joint MRAC systems retain their audited official pipelines and appear in a separate contextual block, not as if their inputs were controlled. HOTA [5], IDF1, and ID switches [7] diagnose predicted tracks. When the same frozen tracks feed several counters, these tracking values must be identical and cannot be credited as counting gains.

Evaluator-ready outputs. Each counter adapter must emit the scored event tuples in Equation 13, the per-track count vector, and the exact spatial support expected by the pinned evaluator. Oracle tracks retain official identity support. Predicted tracks remain separate through evaluation: the manifest-bound adapter must specify one deterministic matching policy and how unmatched, duplicated, fragmented, merged, and identity-switched tracks become false positives or false negatives. No post-hoc fragment joining is allowed unless it is a separately audited method component. Adapter source, tests, and per-instance outputs are required before Period-mAP or person-wise count metrics become table-eligible.

Metrics and uncertainty. Using the frozen official evaluator, we will report Period-mAP, AP50, AP75, AvgMAE, and AvgOBO, together with an identity-conditioned count-error diagnostic after the evaluator's frozen matching step. Period-mAP under common predicted tracks is the confirmatory metric; the primary contrast is our method minus track-conditioned PAMS on identical videos and frontend outputs. A claim of improvement requires the lower endpoint of its paired 95% interval to be above zero. Native-pipeline

rows are descriptive and cannot establish non-inferiority or superiority under a common-input protocol.

Every reproducible method uses at least three pre-registered seeds, reported as mean and between-seed sample standard deviation. Within each seed, the primary paired comparison receives a two-sided 95% percentile-bootstrap interval by resampling videos as clusters and retaining all identities within a sampled video; seed and video-sampling uncertainty are not collapsed into one quantity. Secondary ablations are exploratory, with family-wise Holm correction if formal significance statements are made. Official single-checkpoint numbers remain explicitly marked and are excluded from variance and significance claims. Test annotations are not used for tuning, checkpoint selection, routing calibration, or early stopping.

Baselines and supervision. The mandatory native baselines are MultiCounter and MultiCounter+ [10, 11]. Under the common track interface we adapt RepNet [1], TransRAC [3], PoseRAC [13], and PAMS [2]; DeTRC [6] and D²-STX [12] enter only if their code and output semantics pass the experiment audit. Tables disclose input modality, training labels, track source, and run count. Comparisons across supervision regimes are accuracy–annotation trade-offs, not claims of identical training information.

4.2. Primary Questions

The evaluation is organized around four questions. First, how does a no-count-label pose counter compare on person-wise MRAC under both oracle and predicted tracks? Second, does local routing help beyond a track-conditioned PAMS base when tempo changes within one identity? Third, does response-domain overlap-add reduce boundary errors relative to summing independently decoded windows? Fourth, how do tracking errors, active-person count, and upstream preprocessing affect robustness and efficiency? Each question has a named table or figure and an associated claim that remains withheld until the evidence audit passes. The mechanism study must also compare against running PAMS independently per track, sliding-window PAMS without a learned router, deterministic tempo bins, non-overlapping and center-crop decoding, uniform response averaging, and phase-aligned fusion.

The quantitative interpretation of Table 1 remains withheld until every displayed binding passes the empirical audits.

The ablation interpretation remains withheld until all rows share one frozen protocol and have the required paired records.

The appendix pre-registers router variants, expert counts, local-versus-global tempo, shared batched execution versus independent same-checkpoint replicas, boundary and continuity ablations, tempo/occlusion/identity-switch stress,

cross-dataset transfer, and latency, memory, FLOPs, and throughput as the number of active identities grows. All plots and tables must be generated from frozen JSON, CSV, or Parquet records; manually transcribed values are inadmissible.

5. Conclusion and Limitations

We specify a pose-driven MRAC counter for people whose pace changes within a track. The central design separates identity state while sharing parameters, routes overlapping windows from local period evidence, fuses continuous responses by normalized overlap-add, and decodes each track once. The setting does not remove all annotation and does not jointly optimize perception and counting: the counting objective is intended to avoid person-wise count and period labels, while upstream perception may use external supervision.

This draft has three material limitations. The current local checkout contains only a single-person PAMS reproduction; the multi-person data path, router, continuity loss, and boundary decoder await synchronization with collaborator code. No eligible MultiRep artifact has been received, so the manuscript makes no performance claim. Finally, a modular pose/track pipeline inherits missed detections and identity switches and may scale with the number of active people. The final study must quantify each limitation through oracle/predicted tracks, corruption analysis, and population-scaling measurements. Until those gates close, the paper remains provisional and data-pending.

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Table 1. Planned MultiRep main results. Reproducible rows will show the mean and sample standard deviation over pre-registered seeds; official single-checkpoint rows will be marked. Protocol blocks answer different questions and will not be pooled into one ranking.

Method	Track protocol	Count supervision	P-mAP $\uparrow$	AP50 $\uparrow$	AP75 $\uparrow$	AvgMAE $\downarrow$	AvgOBO $\uparrow$
RepNet	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
TransRAC	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PoseRAC	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PAMS	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
RepNet	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
TransRAC	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PoseRAC	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PAMS	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter	Native	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+	Native	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

Table 2. Required protocol disclosure accompanying the main table. Native rows retain their official pipelines and are not pooled with common-input rows. Every field is manifest-bound.

Method	Result block	Input	Count labels	Period labels	Track source	Runs
RepNet	Common	PENDING	PENDING	PENDING	PENDING	PENDING
TransRAC	Common	PENDING	PENDING	PENDING	PENDING	PENDING
PoseRAC	Common	PENDING	PENDING	PENDING	PENDING	PENDING
PAMS	Common	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	Common	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter	Native context	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+	Native context	PENDING	PENDING	PENDING	PENDING	PENDING

Table 3. Planned additive ablation under oracle tracks. LT: local tempo; SR: soft routing; BS: boundary stitching; TC: track-continuity objective.

Variant	LT	SR	BS	TC	P-mAP $\uparrow$	MAE $\downarrow$
Track-PAMS					PENDING	PENDING
+ LT	✓				PENDING	PENDING
+ SR	✓	✓			PENDING	PENDING
+ BS	✓	✓	✓		PENDING	PENDING
+ TC	✓	✓	✓	✓	PENDING	PENDING

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A. Extended Experimental Contract

This appendix records analyses that are required for the final evidence bundle but may move between main paper and supplementary material after results and the official CVPR 2027 format are frozen.

A.1. Artifact Manifest

Every table-eligible run must provide: model commit; evaluator commit; configuration and preprocessing fingerprints; seed; hardware and software environment; MultiRep release, source, download time, split, and checksums; per-video and per-identity ground truth and predictions; oracle/predicted-track status; logs; and generated-table provenance. An artifact lacking any required field is retained for diagnosis but cannot support a paper claim. UCFRep’s sealed test partition remains untouched until the method, checkpoints, and analysis code are frozen.

A.2. Mechanism Ablations

The mechanism study compares one global tempo per track against window-local tempo; uniform, hard, and soft routing; one, two, three, and five experts; PAMS independently applied to each track; and sliding-window PAMS with deterministic tempo bins. Boundary alternatives include non-overlapping windows, center-crop decoding, uniform response averaging, phase-aligned fusion, normalized overlap-add, and integer-window summation as a diagnostic failure mode. Training is compared with and without the synchronized within-track continuity term. An oracle router based on ground-truth periods may serve as an upper-bound diagnostic only when the required annotation exists, and must never be grouped with deployable methods. Arbitrary person-indexed parameter sets are excluded because identities do not persist across videos; efficiency instead compares shared batched execution with independent replicas of the same frozen checkpoint.

A.3. Robustness and Tracking Diagnostics

Natural tempo variation will be stratified by within-track cycle-duration dispersion. Deterministic count-preserving piecewise time warps, contiguous masking, and injected identity switches will measure degradation from each clean

input using identical corruptions across methods. Perturbations are separated by pipeline level. Pre-tracking video corruptions rerun the frozen frontend and may change HOTA, IDF1, and IDSW; post-track pose/mask corruptions hold tracks fixed, so tracking metrics must remain unchanged. The protocol must record whether an identity switch changes only the track identifier or also the pose content. Results will be stratified by active-person count, asynchronous starts and stops, pauses, occlusion, and track fragmentation.

A.4. Generalization and Efficiency

If protocol-compatible, a model trained on MultiRep will be evaluated without target-domain adaptation on the official RepCount and UCFRep single-person protocols. Dataset-native normalized MAE and OBO will retain their original names rather than being relabeled as MultiRep AvgMAE/AvgOBO. Efficiency will separate detector/pose/tracker preprocessing from counting latency and report parameters, FLOPs, peak memory, latency, counting-only throughput, and full-pipeline throughput as active identities increase. Shared batched execution and independent same-checkpoint replicas are timed separately. If ByteTrack [14] is selected as the common tracker, its exact code and checkpoint revision will be frozen in the manifest; this mention does not pre-select the frontend.

Qualitative figures must be generated directly from frozen per-frame artifacts and include persistent identity colors, valid masks, local period estimates, routing weights, fused responses, and ground-truth/predicted boundaries. At least one failure case is mandatory; curves may not be manually redrawn.

Table 4. Planned mechanism analysis. Parameter counts expose the cost of expert count and weight sharing.

Factor	Setting	P-mAP↑	AvgMAE↓	AvgOBO↑	Params (M)↓
Tempo scope	Track-global	PENDING	PENDING	PENDING	PENDING
	Window-local	PENDING	PENDING	PENDING	PENDING
Router	Uniform	PENDING	PENDING	PENDING	PENDING
	Hard	PENDING	PENDING	PENDING	PENDING
	Soft	PENDING	PENDING	PENDING	PENDING
Boundary	Center crop	PENDING	PENDING	PENDING	PENDING
	Response average	PENDING	PENDING	PENDING	PENDING
	Normalized overlap-add	PENDING	PENDING	PENDING	PENDING

Table 5. Planned robustness report. Parenthesized values will be paired changes from the corresponding clean input with video-cluster bootstrap intervals.

Condition	Level	Severity	P-mAP↑	AvgMAE↓	AvgOBO↑	IDF1↑
Clean	—	—	PENDING	PENDING	PENDING	PENDING
Natural tempo change	Data stratum	Low/medium/high	PENDING	PENDING	PENDING	PENDING
Piecewise time warp	Post-track	PENDING	PENDING	PENDING	PENDING	PENDING
Contiguous occlusion	Pre-track	PENDING	PENDING	PENDING	PENDING	PENDING
Injected ID switches	Post-track	PENDING	PENDING	PENDING	PENDING	PENDING

Table 6. Planned efficiency report under one frozen hardware and timing protocol.

Method	Active IDs	Params (M)↓	GFLOPs↓	Peak GB↓	Preprocess ms/frame↓	Count ms/frame↓	Pipeline FPS↑
MultiCounter	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING